

Experiment 8:

Write a program to implement Linear Regression (LR) algorithm in python

Python Code:

```
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]])
y = np.array([35, 40, 45, 50, 55, 60, 65, 70, 75, 80])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
new_data = np.array([[7]])
predicted_marks = model.predict(new_data)
print("Actual values:", y_test)
print("Predicted values:", y_pred)
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
print("New input (Hours Studied):", new_data[0][0])
print("Predicted Marks:", predicted_marks[0])
```

Output:

```
Actual values: [75 40]
Predicted values: [75. 40.]
Mean Squared Error: 0.0
R2 Score: 1.0
New input (Hours Studied): 7
Predicted Marks: 65.0
```